

Vasudha Kapre, Ph.D.

Postdoctoral Research Assistant
Composites Manufacturing and Simulation Center (CMSC)
School of Mechanical Engineering
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SUMMARY

- Postdoctoral Researcher with expertise in **composite materials, additive manufacturing, and computational modeling**.
- Ph.D. in Aerospace Engineering with six years of research experience in characterization, process–structure–property relationships, and simulation frameworks (DEM–FEM, FEA) for process modeling of carbon fiber reinforced polymers.
- Experienced in **cross-disciplinary collaboration, teaching, and mentoring**, with publications in Polymer Composites and leading conference proceedings.
- Interested in advancing **research and development, teaching, and product innovation** in composites and advanced manufacturing.

EDUCATION

Ph.D., Aerospace Engineering West Lafayette, IN
Purdue University, GPA 3.92/4.00 2020 - 2024

Advisor: Prof. R Byron Pipes

Thesis: Fiber length attrition of long-discontinuous fiber reinforced polymer pellets in a single screw extruder.

M.S, Aerospace Engineering West Lafayette, IN
Purdue University, GPA 3.68/4.00 2018 - 2020
Structures (Major), Materials (Minor)

B. Tech., Civil Engineering Hyderabad, India
Indian Institute of Technology, GPA 8.88/10 2014 - 2018

SKILLS

Modeling & Simulation: ABAQUS, LIGGGHTS, ANSYS, Additive-3D, MATLAB, Python, Paraview

Characterization: X-ray CT, Microscopy, DSC, TGA, DMA, Tensile/Compression/Shear Testing

Microstructural & Shape Analysis: VG Studio, LEICA, ZEISS, ImageJ, Faro ARM Laser Scanner

Other: Wolfram Mathematica, SolidWorks, VIC-3D, GeomagicWrap, IntelliMax, TA Universal Analysis

Collaboration & Communication: Technical writing and presentation, cross-disciplinary teamwork

Languages: English (Proficient), Hindi (Proficient), Telugu (Proficient), Marathi (Basic), Sanskrit (Basic)

RESEARCH EXPERIENCE

Post-doctoral Researcher, Mechanical Engineering, Purdue University
P.I: Dr. Eduardo Barocio

2025

- Execute an industry funded project with Juggerbot3D to develop additive manufacturing capabilities for fiber reinforced thermoset resin systems.
- Support the Composites Additive Manufacturing and Simulation Consortium (CAMS) at the Composites Manufacturing and Simulation Center through simulation, characterization, and validation for printing with various thermoplastic composites.
- Prepare reports and publications that will be submitted to archival journals, contributing to the preparation of funding proposals.
- Actively expanding expertise in AI for process optimization, and machine learning applications for enabling digital twin manufacturing and reducing experimental efforts

PhD Thesis: Fiber length attrition of long-discontinuous fiber (LDF) reinforced polymer pellets in a single screw extruder

2020 – 2024

- Modeled the flow of cylindrical pellets in a single screw extruder using LIGGGHTS, a discrete element method opensource code implement in C++, and used Paraview for pellet trajectory visualization.
- Validated pellet motion in the solid-conveying zone of a single screw extruder for starve-fed extrusion of 40% CF-PPS by building an extruder system with acrylic barrel for visual pellet tracking.
- Created a sequentially coupled discrete element – finite element (DEM-FEM) framework to capture heat-transfer and melting of a single pellet in ABAQUS based on contact information obtained from LIGGGHTS.
- Extended melting and crystallization kinetics model of semi-crystalline 40% CF-PPS and implemented in ABAQUS heat-transfer simulations through user-subroutine UMATHT.
- Developed analytical models based on beam theory for partially molten pellet deformation and fiber-breakage in the melting zone of a single screw extruder.

Research Assistant, Composites Manufacturing and Simulation Center (CMSC)

2019 – 2023

- Collaborated with the additive manufacturing team to develop material cards for short-fiber reinforced polymer composites such as 50% CF-PPS, 25% CF-PESU, and 25% CF-PSU to enable simulations.
- Performed various experiments to characterize the thermo-mechanical and microstructural properties of orthotropic viscoelastic materials used in extrusion deposition additive manufacturing (EDAM)
- Wrote MATLAB codes to measure and obtain the fiber-orientation tensor and fiber-length distribution from optical micrographs using the image-processing toolbox.
- Validated residual stresses and deformation of additively manufactured geometries using Additive-3D, an in-house physics-based simulation software.
- Investigated the effect of printing parameters such as bead aspect ratio, layer height, and stretching ratio on the effective mechanical properties of printed parts and validated.

TEACHING EXPERIENCE

Teaching Assistant, Purdue University

- Analytical Geometry & Calculus – I, MA 16500 Fall 2018
- Elasticity for Aerospace engineers, AAE 55300 Fall 2023
- Mechanics of Composite Materials, AAE 55500 Spring 2024
- Advanced Manufacturing of Composite Materials, AAE 59000 Spring '23, '24
- Aeromechanics II, AAE 20400 Fall 2024
- Mechanics of Materials, ME 32300 Fall 2025

Delivered two lectures for *ME 323: Mechanics of Materials*, introducing new concepts, solving example problems in class, and conducting quizzes. Served as a Teaching Assistant for three Aerospace Engineering courses (*AAE 553, AAE 555, and AAE 204*), where I held office hours, graded assignments, and proctored exams. Assisted in laboratory instruction for *AAE 590: Advanced Manufacturing of Composite Materials* by setting up experiments and guiding students through composite manufacturing processes. Additionally, taught three recitation sections for *MA 165: Analytical Geometry and Calculus I* in the Mathematics Department, engaging first-year undergraduate students through problem-solving and discussion.

PUBLICATIONS AND PRESENTATIONS

1. **V. Kapre**, E. Barocio, and R. B. Pipes, "Single screw extrusion of long discontinuous fiber-reinforced polymers: Pellet motion and heat transfer," *Polymer Composites* **46**, 10102–10113 (2025).
2. **V. Kapre**, E. Barocio, and R. B. Pipes, "Modeling Flow of Long Discontinuous Fiber Reinforced Polymer Pellets in a Single Screw Extruder," *2024 ASME Aerospace Structures, Structural Dynamics, and Materials Conference*, Seattle, WA, April 2024.
3. E. Barocio, P. Pibulchinda, A. J. Thomas, **V. Kapre**, and A. Franc, "Validated Simulation for Large Scale Additive Manufacturing," *2024 The Composites and Advanced Materials Expo*, Anaheim, CA, Oct. 2024, doi: [10.33599/nasampe/c.22.0064](https://doi.org/10.33599/nasampe/c.22.0064).
4. A. J. Thomas, E. Barocio, **V. Kapre**, P. Pibulchinda, F. N. Nguyen, and R. B. Pipes, "Relationship between flow-controlled fiber orientation and spring-in deformation in extrusion deposition additive manufacturing," *2022 International Solid Freeform Fabrication Symposium*, Austin, TX, July 2022, doi: [10.26153/tsw/44204](https://doi.org/10.26153/tsw/44204).
5. **V. Kapre**, E. Barocio, and R. B. Pipes, "Effects of Bead Deposition Parameters on Mechanical Properties in Extrusion Deposition Additive Manufacturing," *2021 Composites and Advanced Materials Expo*, Dallas, TX, Oct. 2021.
6. **V. Kapre**, E. Barocio, and R. B. Pipes, "Analytical Modeling of Pellet Melting and Fiber Attrition in Single-Screw Extrusion of Long Discontinuous Fiber-Reinforced Polymers," *In preparation*.
7. **V. Kapre** and E. Barocio, "Additive Manufacturing of Fiber Reinforced Thermosets: Progress, Challenges, and Future Perspectives," *In preparation*.
8. **V. Kapre** and E. Barocio, "Framework for Simulation of Additive Manufacturing of Fiber-Reinforced Thermoset Composites," *In preparation*.

FELLOWSHIPS

Institute MCM fellowship, IIT Hyderabad

2017 - 2018

PROFESSIONAL SERVICE

Reviewer - *Polymer Engineering & Science*

2024 - Present

PROFESSIONAL AFFILIATIONS

- American Institute of Aeronautics and Astronautics (AIAA)
- Society for the Advancement of Material and Process Engineering (SAMPE)
- Society of Women Engineers (SWE)
- National Postdoctoral Association (NPA)

COMMUNITY INVOLVEMENT

Council Member, Purdue Postdoctoral Association (PPDA)

2025

REFERENCES

Dr. Eduardo Barocio

Postdoc Principal Investigator

Assistant Professor of Mechanical Engineering

Purdue University

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Dr. R Byron Pipes

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Research Lab

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<https://www.purdue.edu/cmssc/>